

MD NAFIZ GAFFAR

Natural Language Processing | Trustworthy & Explainable AI | Healthcare AI

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RESEARCH PROFILE

MSc Data Intelligence student with a Computer Science background and applied research experience in natural language processing, machine learning, healthcare AI, and large-scale unstructured text analysis. Current work compares statistical NLP and Transformer-based semantic approaches and translates experimental findings into practical data-intelligence systems. Research interests include trustworthy and explainable AI, biomedical NLP, Transformer models, and multimodal intelligent systems.

RESEARCH INTERESTS

Natural Language Processing (NLP) • Trustworthy & Explainable AI (XAI) • Healthcare and Biomedical AI • Transformer Models • Machine Learning • Text Mining • Semantic Representation • Multimodal AI

EDUCATION

MSc Data Intelligence | Canterbury Christ Church University | Canterbury, UK | Sep 2025 – Jan 2027
(Expected)

- Proposed dissertation research area: Trustworthy and Explainable NLP for Healthcare. Supervisor: Dr Merlin Kasirajan.
- Relevant coursework: AI for Data Intelligence; Big Data Database Development; Data Visual Analytics; Data in Action; Information Security Management; Principles of Data Intelligence Programming.

BSc Computer Science and Engineering | American International University-Bangladesh (AIUB) | Dhaka, Bangladesh | Sep 2019 – Feb 2025

- Undergraduate thesis: Federated Learning for Early Liver Disease Detection and Prevention.
- Research foundation in machine learning, privacy-aware distributed learning, and healthcare applications of artificial intelligence.

RESEARCH EXPERIENCE

NLP Research Placement Student | Canterbury Christ Church University | Canterbury, UK | May 2026 – Sep 2026

Supervisor: Dr Merlin Kasirajan

- Conduct applied NLP research using large-scale real-world unstructured text and structured metadata, with emphasis on reproducible data preparation and practical AI-driven data intelligence.
- Develop and compare traditional statistical NLP, latent semantic methods, sentence embeddings, and Transformer-based semantic modelling approaches.
- Evaluate model outputs using quantitative and qualitative criteria including topic quality, semantic coherence, coverage, robustness, interpretability, and usefulness for downstream analysis.
- Build research workflows in Python for preprocessing, feature representation, model experimentation, comparative evaluation, visualisation, and structured result generation.
- Translate research experiments toward an end-user-oriented prototype incorporating database integration, interactive visualisation, and deployment-oriented system development.

SELECTED RESEARCH PROJECT

EvidenceOutcomes-XAI: Interpretable Biomedical NLP for Clinical Outcome Extraction | Independent Research Project | 2026 – Present

- Investigate biomedical NLP methods for extracting and interpreting clinical outcomes from randomized controlled trial abstracts.

- Develop reproducible biomedical-text preprocessing and Transformer-based language-modelling workflows for clinical research text.
- Explore explainability methods for improving transparency and interpretation of model outputs in healthcare and biomedical NLP.
- Technologies: Python, Pandas, Scikit-learn, Hugging Face Transformers, PubMedBERT, NLP, Explainable AI.

Repository: github.com/Nafizgaffarcccu-Hub/EvidenceOutcomes-XAI

TECHNICAL SKILLS

Programming & Data: Python, Pandas, NumPy, Jupyter Notebook, Google Colab

Natural Language Processing: Text preprocessing, TF-IDF, text classification, topic modelling, BERT, Transformer models, sentence embeddings, text mining

Machine Learning: Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machines, K-Nearest Neighbours, Scikit-learn, model evaluation

Explainable AI: SHAP, LIME, counterfactual explanation concepts, model interpretability

Databases & Data Platforms: MongoDB, Elasticsearch, Kibana

Visualisation & Analytics: Matplotlib, KNIME, exploratory data analysis

Development & Research Tools: Git, GitHub, reproducible notebook workflows

RELEVANT PROFESSIONAL EXPERIENCE

MIS Executive | Lakeshore Suites | Bangladesh | 2024 – 2025

- Supported management information, business reporting, dashboard-oriented analysis, and data-driven operational decision support.

CRM | Skytech Solutions Ltd. | Bangladesh | 2024

- Supported customer relationship management processes and structured business information workflows.

ADDITIONAL INFORMATION

Languages: Bengali (native), English

Research Objective: Seeking fully funded PhD opportunities beginning in 2027 in NLP, trustworthy AI, healthcare AI, and multimodal intelligent systems.